

Libraries

- many packages can use the same shared library
- you can have multiple versions of the same library installed
- major package upgrades are installed side by side with their older counterparts
- the files have a `.so` or `.so.version` extension, similar to Windows `.dll` files

static libraries - contained in the package

dynamic libraries - shared

System Library path

contained in `/etc/ld.so.conf`

example - `include /etc/ld.so.conf.d/*.conf*`

can have multiple paths

Linux also refers to trusted libraries in `/lib` and `/usr/lib`

Environment variable

`LD_LIBRARY_PATH`

specifies additional directories to search for libraries

example `export LD_LIBRARY_PATH=/usr/local/testlib:/opt/new/lib`

specifies 2 paths separated by `:`

Errors

example

```
$ gimp
gimp: error while loading shared libraries: libXinerama.so.1: cannot~CA
open shared object file: No such file or directory
```

it could find the library `libXinerama.so.1`

troubleshooting

- check if library installed
- if library is installed, you may need to add it to the Library Path
- Sometimes the librarys `path is hardcoded` in the programs binary file, you can create a symbolic `link` from where the program expects it and the location of the library on your system

example

A program may link to `biglib.so.5` but you have `biglog.so.5.2` installed, so creating a `link` would solve the problem

Library Management

libraries can depend on other libraries

so a library may be there, but its dependencies are not installed

ldd

displays a programs shared libraries

example

```
james@laptop:~$ ldd /usr/bin/date
linux-vdso.so.1 (0x00007f4bfddd3000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007f4bfda00000)
/lib64/ld-linux-x86-64.so.2 (0x00007f4bfddd5000)
```

ldconfig

updates caches and links used by the system for locating libraries
it reads `/etc/ld.so.conf` and implements any changes in that file

Rebuilding Library Cache

Linux doesn't read `etc/ld.so.conf` every time a program runs
instead it relies on a cached stored in `/etc/ld.so.cache`
it runs automatically when you install/uninstall packages

To update the library cache manually, use `ldconfig`

`ldconfig`

	option
-v	display verbose information
-N	don't rebuild cache, but update symbolic links to libraries
-p	display current cache